

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10 – RADIOACTIVITY AND STABILITY OF ISOTOPES (3.1): FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Use everything you have learned from Lesson 1 to Lesson 7 to explain the case of 'The Invisible Killer in the Rock': a piece of pitchblende (uranium ore) placed next to photographic film in complete darkness left a clear exposure on the film. Answer each section in the space under its prompt. Write in full sentences, use correct science words (nucleus, isotope, belt of stability, half-life, alpha, beta, gamma), and write nuclear equations where asked. Always link your ideas back to the rock. Use Kenyan examples from our lessons where you can. Your teacher will mark your work using the rubric.

Part 1: Why is the rock radioactive?

Our driving question asks: 'Why do some atoms in Kenyan rocks spontaneously release invisible energy?' (a) Explain what the nucleus of an atom is made of and what the terms atomic number (Z), mass number (A) and neutron number (N) mean. (b) Explain what an isotope is, using Carbon-12 and Carbon-14 as your example. (c) Using the idea of the belt of stability on an N vs Z graph, explain why the uranium-238 in the pitchblende is different from stable nuclei, and why this is a nuclear process and not a chemical one.

(a) The nucleus is made of protons and neutrons, together called nucleons. The atomic number Z is the number of protons. The mass number A is the total number of protons and neutrons. The neutron number is $N = A - Z$. (b) Isotopes are atoms of the same element with the same Z but different N . Carbon-12 and Carbon-14 both have 6 protons, so they behave the same in chemical reactions, but Carbon-14 has two extra neutrons (8 instead of 6). This different neutron number gives them different nuclear stability: Carbon-12 is stable while Carbon-14 is unstable. (c) Stable nuclides cluster in a band on the N vs Z graph called the belt of stability. Uranium-238 has a neutron-to-proton ratio that places it outside this belt, so its nucleus is unstable. An unstable nucleus spontaneously emits particles or energy, which is called radioactivity. This happens inside the nucleus, so it needs no outside energy and cannot be switched off by ordinary chemical means. That is why the rock keeps emitting radiation and exposed the film in the dark.

Part 2: What is coming out of the rock?

(a) Name the three types of radiation that unstable nuclei can emit. For each, state its charge, its relative penetrating

(a) Alpha (α): charge +2, mass about 4 amu, low penetration, stopped by a sheet of paper. Beta (β): charge -1, mass about 0, moderate penetration, stopped by a few millimetres of aluminium. Gamma (γ): charge 0, high penetration, needs lead (or thick concrete) to reduce it. (b) Alpha particles are large and highly charged, so they strongly ionise the atoms they pass. Each ionisation takes energy from the particle, so it loses its energy quickly and is stopped after a very short distance. High ionisation therefore

power and what stops it. (b) Alpha radiation has the highest ionising ability, yet it is considered the least penetrating. Explain how both can be true. (c) Explain why alpha particles from the ore are still dangerous in a Kenyan mining situation even though paper stops them.	goes with low penetration, and gamma rays show the opposite pattern. (c) Alpha particles are not a big danger from outside the body because paper or the skin's surface stops them. If dust from uranium-bearing rock is inhaled or swallowed, however, the alpha emitter is inside the body in direct contact with tissue, where its strong ionisation can damage cells. So alpha radiation is most hazardous when taken in internally.
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Part 3: Nuclear equations and the stability plot

(a) Write the balanced nuclear equation for the alpha decay of uranium-238 into thorium. (b) Write the nuclear equation for the beta-minus decay of thorium-234 into protactinium. (c) State what happens to A and Z in alpha decay, beta-minus decay and gamma emission. (d) Explain, using the N vs Z graph, how the decays of uranium-238 and its daughters move the nucleus on the plot, and what this shows about the nucleus and the belt of stability.	(a) $^{238}_{92}\text{U} \rightarrow ^{234}_{90}\text{Th} + ^4_2\text{He}$. The top numbers add up ($238 = 234 + 4$) and the bottom numbers add up ($92 = 90 + 2$). (b) $^{234}_{90}\text{Th} \rightarrow ^{234}_{91}\text{Pa} + ^0_{-1}\text{e}$ (a beta particle). A stays 234 and Z rises from 90 to 91; the charges balance ($90 = 91 - 1$). (c) Alpha decay: A decreases by 4 and Z decreases by 2. Beta-minus decay: A stays the same and Z increases by 1. Gamma emission: neither A nor Z changes, only energy is released. (d) On the N vs Z graph, uranium-238 lies outside the belt of stability. Alpha decay reduces both N and Z by 2, so the nucleus moves diagonally down and to the left on the plot. Beta-minus decay lowers N by 1 and raises Z by 1, so the nucleus moves one step diagonally down and to the right. Each decay gives a daughter nucleus plotted in a new position, so the nucleus changes its composition step by step along the chain. Checking the neutron-to-proton ratio shows that not every single step moves it in the same direction: for uranium-238, $N/Z = 146/92 \approx 1.59$, and for thorium-234, $N/Z = 144/90 = 1.60$, so the first alpha step does not clearly move the nucleus toward the belt in ratio terms. The beta step to protactinium-234 gives $N/Z = 143/91 \approx 1.57$, so the ratio falls. The daughters may still be unstable and decay again, and the chain as a whole continues until a stable nucleus is reached. Gamma emission does not move the nucleus on the plot; it carries away excess energy during these changes.
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Part 4: Detecting the radiation and how long it lasts

(a) Explain how a Geiger-Müller counter turns invisible radiation from the rock into a click. (b) A GM counter clicks even when no source is nearby. Explain why, and how you should handle this when measuring the rock. (c) Define half-life. Explain, using your dice simulation and the half-life of uranium-238, why the pitchblende rock has been emitting radiation for billions of	(a) Radiation enters the GM tube through a thin mica window and ionises argon gas atoms inside. The freed electrons accelerate toward the central anode under high voltage, producing an electrical pulse that the counter registers as a click. (b) Background radiation is always present from cosmic rays, radon gas from soil, naturally radioactive building stone such as Rift Valley granite, and trace radioisotopes in food such as potassium-40 in sukuma wiki and githeri. When measuring the rock, you must record the background count rate first and compare it with the reading near the rock, to find the true emission from the source. (c) Half-life is the time for half of the undecayed nuclei in a sample to decay. In the dice simulation, the number of surviving dice fell each round along a curve that flattened out, and fewer dice were removed each round as fewer remained. Decay of any single nucleus is random, but for a very large number of nuclei it is statistically predictable. Uranium-238 has a half-life of 4.5 billion years, which is about the age of the Earth, so about half of the original uranium-238 is still undecayed. A huge number of undecayed nuclei remain, and the rate of emission falls only very slowly, so the rock still emits radiation and appears to do so indefinitely on a human timescale. (d) 24 days is $24 \div 8 = 3$ half-lives. $N = N_0 \times (1/2)^3 = N_0 \times 1/8$. One-eighth of the nuclei remains.
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years. (d) A sample of Iodine-131 used at KNH has a half-life of 8 days. Using $N = N_0 \times (1/2)^n$, what fraction of the nuclei remains after 24 days?	
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Part 5: Closing the case: energy, uses and safety

(a) Explain where the energy released from the pitchblende ultimately comes from, using the idea of mass defect. (b) Describe the difference between nuclear fission and nuclear fusion. (c) Give two Kenyan applications of radioactivity from our lessons and explain how each uses a property of radiation or half-life. (d) Explain the ALARA principle and describe three ways to reduce exposure to radiation. (e) Write a short explanation, for a Year 9 student, of why some rocks are radioactive and why it matters for Kenya.	(a) In nuclear changes the products have slightly less mass than the starting nucleus. This missing mass, the mass defect, is converted into energy according to $E = mc^2$. So the energy from the rock comes from the mass of the unstable uranium nuclei themselves. (b) Fission is the splitting of a heavy nucleus such as uranium-235, struck by a neutron, into smaller fragments plus more neutrons and a large amount of energy. Fusion is the joining of light nuclei, such as hydrogen isotopes, under extreme conditions, releasing energy. (c) Example 1: Cobalt-60 radiotherapy at Kenyatta National Hospital uses penetrating gamma rays to reach and destroy cancer cells in patients such as those with cervical cancer. Example 2: KALRO gamma irradiation of stored maize, beans and sorghum uses radiation to kill pests. Other acceptable examples: carbon-14 dating of Olorgesailie charcoal using its half-life of 5730 years, or Iodine-131 thyroid treatment at KNH using its short half-life of 8 days. (d) ALARA means keeping exposure As Low As Reasonably Achievable. Three ways: use shielding such as lead or concrete; increase distance, because doubling the distance reduces the dose to one-quarter; and minimise the time spent near the source. (e) Some rocks, like pitchblende, contain atoms such as uranium-238 whose nuclei have a neutron-to-proton ratio outside the belt of stability, so they are unstable. Uranium-238 spontaneously decays by emitting alpha particles, and the daughter nuclei in the decay chain then emit beta particles and gamma rays as the chain continues. The energy comes from a tiny loss of mass. Because the half-life is 4.5 billion years, a very large number of undecayed nuclei remain and the rock seems to emit forever. This matters for Kenya because the same radiation can harm people if not controlled, but with detection and safety rules it can treat cancer, protect food and help date fossils such as those at Koobi Fora.
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RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining why the nucleus is unstable	Accurately explains nucleons, isotopes and the belt of stability, and clearly links the neutron-to-proton ratio of uranium-238 to spontaneous decay as a nuclear, not chemical, process.	Explains isotopes and instability correctly but the link to the belt of stability or to the nuclear nature of decay is incomplete.	Describes radioactivity without explaining nuclear instability or confuses isotopes with other ideas.
Radiation types and nuclear equations	Correctly compares alpha, beta and gamma, writes balanced equations with conserved A and Z, and explains decay arrows on the N vs Z plot.	Compares radiation types correctly and writes mostly correct equations with minor errors in balancing or in explaining movement on the plot.	Lists some properties but equations are unbalanced or missing, and no link to stability is made.
Detection, background radiation and half-life	Explains the GM tube process, background correction and half-life using the dice evidence, and correctly	Explains most of detection and half-life correctly; the calculation or the use of simulation evidence has small gaps.	Gives vague statements about detection or half-life; calculation is missing or incorrect.

	calculates the fraction remaining.		
Energy source, applications and safety	Links mass defect and $E = mc^2$ to the rock's energy, distinguishes fission from fusion, and gives Kenyan applications and ALARA measures with correct reasoning.	Gives correct applications and safety measures, with only partial explanation of mass defect or of fission versus fusion.	Names applications or safety ideas without explanation, or with errors.
Communication and connection to the phenomenon	Writes clearly in logical order, uses scientific vocabulary precisely, and consistently ties each idea back to the pitchblende rock and Kenyan contexts.	Writing is mostly clear and uses key terms; links to the phenomenon are present but not consistent.	Writing is hard to follow, with little scientific vocabulary or connection to the rock.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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